

Compact Combustion Diagnostics Rig

Independent Design Project — Sample Report

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OVERVIEW

This sample report summarizes a bench-scale combustion diagnostics rig designed to characterize flame behaviour under varying fuel-air ratios. The project combined mechanical design, sensor integration, and post-processing software.

APPROACH

A modular burner assembly was designed in SolidWorks and manufactured from stainless steel components. Optical access ports were added to support laser-based imaging. A MATLAB pipeline was built to process high-speed camera data into flame-front metrics.

RESULTS

The rig achieved repeatable flame stabilization across a target range of equivalence ratios, and reduced experiment setup time by roughly a third compared to the prior bench configuration.

NOTES

This document is a placeholder used to demonstrate the portfolio site's in-page, view-only document viewer. Replace with a real project write-up.